Artificial Intelligence (AI-2002)

Assignment 2: Analytical Report

Constraint Satisfaction, Adversarial Search and Probability

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This report accompanies the source notebook *AI2002_Assignment2_24K-2523.ipynb*. It contains the formulations, heuristic designs, analysis, mathematical justifications and verification logs for Parts A, B and C.

Part A: Constraint Satisfaction Problem

A.1 Formulation

Variables: BA, SD, PB, KP, GB, AK, ICT. **Domains:** {Red, Green, Blue} for each. **Constraints:** colour(X) != colour(Y) for every pair of bordering regions. The adjacency list is symmetric (verified by an assertion in the code). The constraint graph has 7 nodes and 11 edges.

Region	Neighbours	Degree
BA	SD, PB, KP	3
SD	BA, PB	2
PB	BA, SD, KP, ICT, AK	5
KP	BA, PB, ICT, GB, AK	5
GB	KP, AK	2
AK	GB, KP, PB	3
ICT	KP, PB	2

A.2 Algorithms

Basic backtracking: depth-first search that picks unassigned variables in alphabetical order (AK, BA, GB, ICT, KP, PB, SD) and values in the order Red, Green, Blue. A value is accepted only if it conflicts with no already assigned neighbour.

Enhanced solver: (1) **MRV** selects the variable with the fewest legal colours left. (2) The **Degree heuristic** breaks MRV ties by choosing the variable with the most unassigned neighbours. (3) **Forward checking** removes the chosen colour from the domains of unassigned neighbours after every assignment and backtracks immediately if any domain becomes empty. Removed values are restored on backtrack.

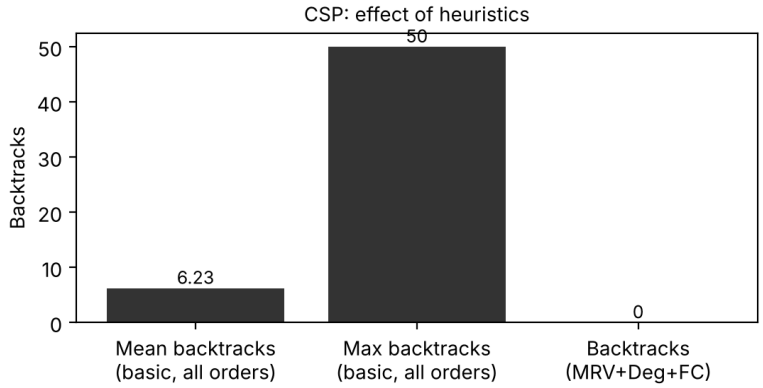
A.3 Performance analysis

Metric (default alphabetical order)	Basic backtracking	MRV + Degree + FC
Recursive calls	8	8
Assignments tried	7	7
Backtracks	0	0
Valid solution	Yes	Yes

Both solvers solve the map with the minimum possible 7 assignments and 0 backtracks under the default order. The basic solver is lucky here because the alphabetical order happens to avoid dead ends, and the map is small. To compare fairly, the basic solver was run on **all 5040 variable orderings**:

Basic solver, all 5040 orders	Mean	Max
Backtracks	6.23	50
Assignments	13.23	57
Orders with at least 1 backtrack	3070 of 5040	

The heuristic solver always gives 7 assignments and 0 backtracks. About 61% of the orderings make the basic solver backtrack, and the worst one needs 50 backtracks. MRV and Degree start with the most constrained regions (KP and PB have 5 neighbours each) and forward checking detects a dead end as soon as a domain is empty. The benefit would grow quickly on larger maps.



Solutions found. Basic: AK=Red, BA=Red, GB=Green, ICT=Red, KP=Blue, PB=Green, SD=Blue. Heuristic: KP=Red, PB=Green, AK=Blue, BA=Blue, GB=Green, ICT=Blue, SD=Red. Both were checked against every constraint.

Part B: Adversarial Search (Isolation, 5 x 5)

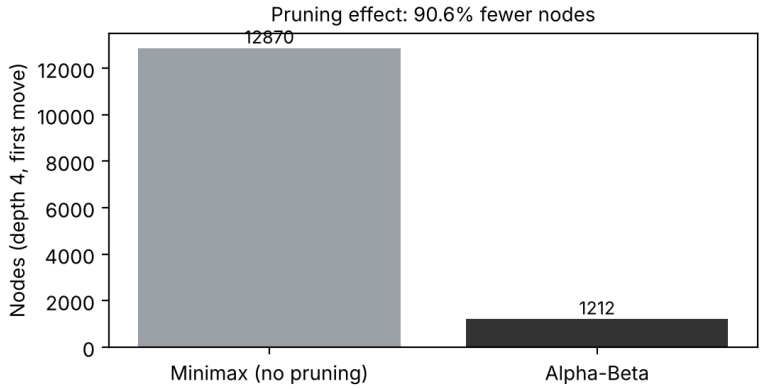
B.1 Game model and engine

The board is 5 x 5. The AI starts at (0,0) and the human at (4,4). Moves follow the **Queen** trajectory (any distance in 8 directions, cannot cross or land on occupied or destroyed cells). The cell a player vacates is destroyed permanently. A player with no legal move on their turn loses.

Search. The AI is the MAX player and the human is the MIN player. Alpha-beta pruning is used with a ceiling of **4 plies**. At a terminal node the value is +infinity if the human cannot move and -infinity if the AI cannot move. At depth 0 a heuristic scores the leaf. The cutoff condition is **alpha >= beta** (checked at both MAX and MIN nodes), after which the remaining children are skipped.

Mathematical justification. The AI has 11 legal moves at the start, so plain minimax to depth 4 explores a tree of thousands of nodes (12,870 in our measurement). For a tree of branching factor b and depth d, minimax visits $O(b^d)$ nodes. With good move ordering alpha-beta visits about $O(b^{d/2})$ nodes, which lets the same time budget search roughly twice as deep. Pruning is safe because once beta <= alpha, the MIN player already has an option that is no better for MAX than something MAX can guarantee elsewhere, so the value of the remaining branches cannot change the root decision.

Measurement (first AI move, depth 4)	Value
Plain minimax nodes	12,870
Alpha-beta nodes	1,212
Cutoffs triggered	226
Reduction	90.6%



B.2 Heuristic evaluation functions

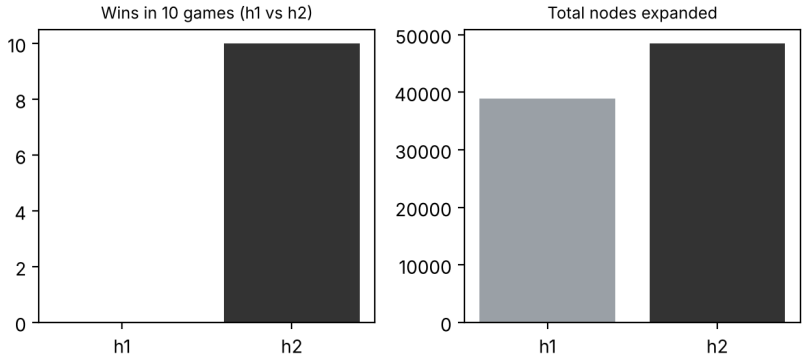
Function 1 (h1), mobility differential: h1 = (number of AI moves) - (number of human moves).

Function 2 (h2), territory and buffer score: h2 = 2.0 x (territory difference) + (my moves - 1.5 x opponent moves) + 0.5 x (centre bonus). Territory uses a Queen-move BFS from both players: a cell belongs to the player who reaches it in fewer moves, and ties give a half point to the player to move. The 1.5 weight rewards restricting the opponent. The centre bonus is the Manhattan closeness to the middle of the board, where more routes exist.

B.3 Comparison of the two heuristics

Aspect	h1 (mobility)	h2 (territory + buffer)
Information used	Immediate moves only	Reachable territory, mobility, position
Cost per leaf	Very low	Higher (two BFS per leaf)

Aspect	h1 (mobility)	h2 (territory + buffer)
Strength	Opening, close combat	Mid and end game, detects when a player is cut off
Weakness	Cannot see closed-off regions or traps	More computation per node
Result vs random (20 games)	20 wins	20 wins
Head-to-head (10 games, sides swapped)	0 wins	10 wins
Total nodes expanded	38,930	48,475



Discussion. Both heuristics beat a random player every time, so both are sound. In direct play h2 won all 10 games, with each side starting equally often as AI. h1 sees only the next step, so two positions with the same move count look equal even when one leads into a small dead region. h2 measures which cells each player can reach first, so it recognises a partition of the board early and steers toward the larger region. The price is about 25% more nodes expanded per game, because each leaf needs two BFS runs. The average game length was 19.5 plies.

Part C: Probability by Enumeration

The joint distribution factorises along the network: P(Re,At,Bu,Ov,Cr,Al,Co,Es) = P(Re) P(At) P(Bu|Re) P(Ov|At) P(Cr|Bu,Ov) P(Al|Cr,At) P(Co|Cr) P(Es|Al,Co). Queries use P(Q|e) = alpha x sum over hidden variables of P(Q,e,h). Notation: T means true, F means false, and a leading ~ means not.

(a) P(Cr) [4]

P(Bu) = 0.2(0.3) + 0.8(0.05) = 0.10. P(Ov) = 0.1(0.6) + 0.9(0.1) = 0.15. Bu and Ov are independent because their parents are independent.

Bu	Ov	P(Bu)P(Ov)	P(Cr|Bu,Ov)	Product
T	T	0.0150	0.95	0.01425
T	F	0.0850	0.60	0.05100
F	T	0.1350	0.40	0.05400
F	F	0.7650	0.03	0.02295

P(Cr) = 0.01425 + 0.05100 + 0.05400 + 0.02295 = **0.1422**

(b) P(Al) [4]

P(Cr|Ov) = 0.1(0.95) + 0.9(0.4) = 0.455 and P(Cr|~Ov) = 0.1(0.6) + 0.9(0.03) = 0.087. So P(Cr|At) = 0.6(0.455) + 0.4(0.087) = 0.3078 and P(Cr|~At) = 0.1(0.455) + 0.9(0.087) = 0.1238.

At	Cr	P(At,Cr)	P(Al|Cr,At)	Product
T	T	0.03078	0.99	0.0304722
T	F	0.06922	0.70	0.0484540
F	T	0.11142	0.90	0.1002780
F	F	0.78858	0.05	0.0394290

P(Al) = 0.2186332, which is **0.2186** to 4 decimal places.

(c) P(Cr | Re) [4]

Given Re, P(Bu) = 0.3 and P(Ov) = 0.15.

Bu	Ov	P(Bu|Re)P(Ov)	P(Cr|Bu,Ov)	Product
T	T	0.045	0.95	0.04275
T	F	0.255	0.60	0.15300
F	T	0.105	0.40	0.04200
F	F	0.595	0.03	0.01785

P(Cr|Re) = 0.04275 + 0.15300 + 0.04200 + 0.01785 = **0.2556**

(d) P(Es) [6]

P(Es) = sum over Al,Co of P(Al,Co) P(Es|Al,Co), where P(Al,Co) = sum over At,Cr of P(At,Cr) P(Al|Cr,At) P(Co|Cr), with P(At,Cr) from part (b).

Al	Co	P(Al,Co)	P(Es|Al,Co)	Product
T	T	0.119926	0.95	0.113930
T	F	0.098707	0.60	0.059224
F	T	0.086724	0.40	0.034690
F	F	0.694643	0.01	0.006946

P(Es) = 0.113930 + 0.059224 + 0.034690 + 0.006946 = 0.214790, which is **0.2148**.

(e) P(At | Al, ~Co) [6]

Each term is P(At,Cr) x P(Al|Cr,At) x P(~Co|Cr).

At	Cr	Calculation	Value
T	T	0.03078 x 0.99 x 0.15	0.004571
T	F	0.06922 x 0.70 x 0.90	0.043609
F	T	0.11142 x 0.90 x 0.15	0.015042
F	F	0.78858 x 0.05 x 0.90	0.035486

Numerator P(At,Al,~Co) = 0.004571 + 0.043609 = 0.048179. Denominator P(Al,~Co) = 0.048179 + 0.015042 + 0.035486 = 0.098707. So P(At|Al,~Co) = 0.048179 / 0.098707 = **0.4881**.

(f) P(Re | Cr, ~Al) [6]

Denominator: P(Cr,~Al) = P(At,Cr)(0.01) + P(~At,Cr)(0.10) = 0.0003078 + 0.011142 = 0.0114498.

Numerator: with Re true, P(Cr|Re,Ov) = 0.3(0.95) + 0.7(0.4) = 0.565 and P(Cr|Re,~Ov) = 0.3(0.6) + 0.7(0.03) = 0.201. Then P(Cr|Re,At) = 0.6(0.565) + 0.4(0.201) = 0.4194 and P(Cr|Re,~At) = 0.1(0.565) + 0.9(0.201) = 0.2374. So P(Re,At,Cr) = 0.2(0.1)(0.4194) = 0.008388 and P(Re,~At,Cr) = 0.2(0.9)(0.2374) = 0.042732. P(Re,Cr,~Al) = 0.008388(0.01) + 0.042732(0.10) = 0.00435708.

P(Re|Cr,~Al) = 0.00435708 / 0.0114498 = **0.3805**.

Summary of answers

Query	Answer
(a) P(Cr)	0.1422
(b) P(Al)	0.2186
(c) P(Cr | Re)	0.2556
(d) P(Es)	0.2148
(e) P(At | Al, ~Co)	0.4881
(f) P(Re | Cr, ~Al)	0.3805

Verification Logs

The following output was produced by running the submitted notebook. All six probabilities were also computed by brute-force enumeration over all $2^8 = 256$ worlds, and they agree with the hand calculations above. The joint distribution was asserted to sum to 1.

CSP

```
Basic    : calls=8 assignments=7 backtracks=0 valid=True
Heuristic: calls=8 assignments=7 backtracks=0 valid=True
Basic over all 5040 orders: backtracks mean=6.23 max=50;
  orders with >=1 backtrack = 3070; assignments mean=13.23 max=57
```

Isolation

```
AI legal moves at start: 11
h1 vs random: 20/20 wins     h2 vs random: 20/20 wins
Head-to-head (10 games): h1=0, h2=10
Nodes expanded: h1=38930, h2=48475; avg game length = 19.5 plies
Minimax nodes: 12870 | Alpha-Beta nodes: 1212 | Cutoffs: 226
Reduction: 90.6%
```

Probability

```
Joint sums to 1: OK
P(Cr)           = 0.1422
P(Al)           = 0.2186
P(Cr | Re)      = 0.2556
P(Es)           = 0.2148
P(At | Al, ~Co) = 0.4881
P(Re | Cr, ~Al) = 0.3805
```